

End Semester Examination: November-2019
B.E. EC-Third Semester
ECC08 SIGNALS AND SYSTEMS

Time duration: 3 hours

Total Marks: 50

Note: Attempt **10** questions. All questions carry equal marks. Assume suitable missing data, if any.

1. For the systems $y(n) = x(n^2)$ and $y(t) = \int_{-\infty}^{2t} x(\vartheta) d\vartheta$, determine whether they are (i) memoryless (ii) causal (iii) time-invariant (iv) linear.
2. Consider the signal $b(n) = x(n) - \alpha x(n-1)$ where $x(n) = \alpha^n u(n)$. Using these signals, obtain the sequence $h(n)$ such that $x(n) * h(n) = \left(\frac{1}{2}\right)^n [u(n+2) - u(n-2)]$.
3. Determine the Fourier transform of the signal $m(t) = te^{-|t|}$. Using this result, also determine the Fourier transform of the signal $p(t) = \frac{4t}{(1+t^2)^2}$.
4. For the following system functions of the LTI systems, obtain the corresponding z -transforms and also comment on the causality of these systems.

$$(a) H_1(z) = \frac{3 - 4z^{-1}}{1 - 3.5z^{-1} + 1.5z^{-2}}, |z| < 0.5$$

$$(b) H_2(z) = \frac{-1 - 0.4z^{-1}}{1 - 2.8z^{-1} + 1.6z^{-2}}, 0.8 < |z| < 2$$

5. Consider an LTI system that is characterized by the differential equation

$$\frac{d^2 y(t)}{dt^2} + 4 \frac{dy(t)}{dt} + 3y(t) = \frac{dx(t)}{dt} + 2x(t)$$

Determine the response if the input is $x(t) = e^{-t}u(t)$. Also, obtain $y(t)$ and comment on the causality and stability of this system.

6. Consider a periodic signal $x(n)$ with $N = 8$ and $X_k = -X_{k-4}$. Obtain the function $G(k)$ such that $Y_k = G(k)X_k$, where the signal $y(n)$ is given as

$$y(n) = \left[\frac{1 + (-1)^n}{2} \right] x(n-1).$$

Here X_k and Y_k denote the Fourier series coefficients of $x(n)$ and $y(n)$ respectively.

7. Following information is provided about the signal $x(t)$ with Laplace transform $X(s)$.
i) $x(t)$ is real.

- ii) $X(s)$ has two poles and no zeros in the finite s-plane.
- iii) $X(0) = 8$.
- iv) $X(s)$ has a pole at $s = -1 + j$.
- v) $e^{2t}x(t)$ is not absolutely integrable.
- vi) $X(1) = 4$.

Determine $X(s)$ and $x(t)$. Also, specify ROC of $X(s)$.

8. Determine the Hilbert transform of $x(t) = \frac{1}{1+t^2}$. Prove that a signal and its Hilbert transform will always have same energies and are orthogonal in nature.
9. The response of a system to the input $(n+2)\left(\frac{1}{2}\right)^n u(n)$ is obtained as $\left(\frac{1}{4}\right)^n u(n)$. If the response of this system changes to $\delta(n) - \left(\frac{-1}{2}\right)^n u(n)$, then what will be the input?
10. Explain the phenomenon of reconstruction of an information signal from its samples using the method of interpolation with suitable diagram. Consider the signal $x(t) = (3\cos(2000\pi t) + 5\sin(6000\pi t) + 10\cos(12000\pi t))u(t)$. Assume that the sampling of this signal is achieved at a sampling rate of $f_s = 5000$ samples per second. Determine the discrete time signal obtained after sampling. Find the z-transform of this signal along its ROC.

11. Describe in detail the ROC and its properties in relation to Laplace transform.

12. Consider the signal

$$x(t) = \sum_{k=-\infty}^{\infty} \left(\frac{\sin(k\pi/4)}{k\pi/4} \right) \delta(t - k\pi/4)$$

Obtain $y(t)$ such that $x(t) = \left(\frac{\sin(t)}{\pi t} \right) y(t)$. Also, determine $X(\omega)$ and comment on its periodicity.

13. Consider a discrete time LTI system with frequency response

$$H(e^{jw}) = \left[\frac{1 + e^{-2jw} + 4e^{-4jw}}{1 + 0.5e^{-2jw}} \right] e^{-j(w - \frac{\pi}{4})}$$

for $-\pi < w < \pi$. Determine the discrete time Fourier transform of the output if the input is $x(n) = \cos\left(\frac{\pi n}{2}\right)$.

14. For the periodic signal

$$x(t) = 2 + \cos\left(\frac{2\pi t}{3}\right) + 4\sin\left(\frac{5\pi t}{3}\right),$$

obtain the fundamental frequency ω_0 and the Fourier series coefficients X_n such that

$$x(t) = \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t}$$